

RCA ref 17653-101/1



25 July 2025

City of Newcastle
[REDACTED]

Attention: [REDACTED]

Geotechnical Engineering

Engineering Geology

Environmental Engineering

Hydrogeology

Construction Materials Testing

Environmental Monitoring

Noise & Vibration

Occupational Hygiene

PROPOSAL FOR GEOTECHNICAL INVESTIGATION NEW LAMBTON LANDSLIDE

1 INTRODUCTION

Thank you for your email inviting us to submit a proposal for geotechnical studies at the May 2025 New Lambton landslide site.

RCA presented preliminary assessment of the landslide in June 2025 (RCA 17653-201r0 Preliminary Report, June 2025) and recommended a detailed geotechnical investigation and outlined an indicative scope. Subsequent to delivery of that report, RCA have inspected the site and held meetings with City of Newcastle (CoN) and Transport for NSW representatives to discuss requirements for the investigation and outcomes.

RCA understand that CoN require a geotechnical investigation of the landslide to provide an assessment of the nature and extent of the landslide and to provide analysis of the landslide, along with discussion of concept options for stabilisation with specific focus on the public assets (Russell Road, Victoria Street and Baker Street).

The scope also includes assessment of potential impacts of abandoned coal mines that lie beneath the site at a depth of between about 25-50m.

This proposal provides a minimum recommended scope of investigation and a costing for that scope. The scope provided is considered to represent the minimum suggested scope based on specific focus on public assets and is considered likely to expand once an understanding of the site conditions and instability mechanism is obtained and to provide inputs into detailed design of stabilisation options.

The proposal has been prepared based on current Contract 2024/076T (Geotech Panel).

2 PROJECT UNDERSTANDING

Project understanding and requirements are detailed in RCA 17653-201r0 Preliminary Report, June 2025 and reference should be made to that report.

3 PROPOSED SCOPE OF WORKS

Based on the site details and the expected geotechnical conditions at the site, we propose the following investigation methodology.

3.1 METHODOLOGY

The scope is set out in **Drawing 1** attached which includes 10 nominated bore locations set out above the identified landslide area and down the landslide area towards its base. Locations are approximate and subject to access approvals, service clearance and assessment of accessibility for equipment and personnel. Locations are numbered 1-10 on **Drawing 1**. Allowance has been made for drilling and instrumentation as listed in **Table 1**. At each of the 10 locations one bore will be cored into rock with recovery of rock core and allowance for downhole imaging and analysis. At 6 locations, allowance is made for installation of an inclinometer and a slotted standpipe piezometer in a second open hole bore at that location.

Drawing 2 provides a preliminary mine overlay plan of the site identifying the bord and pillar working layout under the site in the Borehole Coal Seam. Two bores (Bores 5 and 8) will be extended to the floor of the Borehole Seam to assess the depth and condition of the mine overburden strata and coal pillars. Allowance is made for downhole camera inspection of these two extended bores and sonar imaging if large voids are encountered.

Assumed depths of bores are listed in **Table 1**.

Table 1 *Proposed Bore and Instrumentation Depths*

Bore Number	Assumed Depth (m)	Cored Bore	Inclinometer	GW Bore (open hole)	Bore to Borehole Coal Seam floor	Land Ownership
1	25	Yes	No	No	No	Public
2	25	Yes	Yes	Yes	No	Public
3	25	Yes	No	No	No	Public
4	20	Yes	Yes	Yes	No	Private
5	50	Yes	Yes (top 20m)	Yes	Yes	Public
6	20	Yes	No	No	No	Private
7	15	Yes	No	No	No	Private
8	40	Yes	Yes (top 15m)	Yes	Yes	Public
9	15	Yes	Yes	Yes	No	Public
10	10	Yes	Yes	Yes	No	Public
SUM	245	-	105	105	-	-

3.2 FIELDWORK

- Boreholes drilled to indicative depths listed in **Table 1** using a drilling rig. Drilling will be undertaken using continuous spiral flight augers and rock coring methods for cored bores and PCD open hole drilling techniques would be used in the paired groundwater monitoring bores.
- In the cored bores standard penetration tests would be undertaken at regular intervals to assess soil strength and thin wall samples (U63 or U75 tubes) would be taken in clay soils.
- Rock coring would be undertaken at the 10 bore locations where rock occurs within the target depth. Rock core will be recovered with point load strength testing and photography of the core.
- Installation of inclinometers in 6 cored bores. The inclinometers will comprise non-slotted inclinometer casing and will be founded in a minimum of 1.5m of 'sound' rock. Where feasible inclinometers will be finished with 300-400mm PVC stick up with a standpipe monument cover, locked cap and note of the inclinometer number, job number and depth of lowest reading. The direction in which movement is expected will be marked the "A" direction. Where stick up is not feasible, inclinometers will be finished below ground level with lockable flush capping covers.
- Installation of 50mm diameter standpipe piezometers at 6 twin bore locations to allow for measurement of groundwater levels / sampling of the groundwater/in situ permeability testing.
 - RCA has allowed to drill to depths listed in **Table 1** on the presumption that groundwater will be encountered within those depths.
 - All groundwater bores will be installed by a licensed drilling contractor.
 - All groundwater monitoring wells will be constructed from Class 18 PVC, with machine slotted screens (3m) and screw type fittings. Bores will be backfilled using washed river gravel to the required depth and sealed using a bentonite seal. PVC casing will either be left protruding from the ground by approximately 0.5m stick up to enable protection by a standpipe monument cover or terminated just below existing surface and protected with a road box installed flush with the ground surface.
 - At the completion of drilling, each groundwater bore will be developed to remove disturbed fines and introduced water.
 - Allowance is made to install a Hobo water level data logger in each of the 6 standpipe piezometers to record groundwater levels over time, together with a further data logger to record barometric pressures over time at the site.
 - No allowance is made for groundwater bores targeted at particular seams or strata. Allowance is only made for standpipe piezometers sealed at the surface.
- Allowance has been made for imaging (optical and/or acoustic) of the cored bores to assess rock structure.
- Allowance is made for downhole camera inspection of the two mine bores and sonar imaging if significant voids are encountered at mine level.

- Allowance is made for excavation of test pits by tracked excavator at selected locations where it is suspected the potential slide plane is within reach of the excavator. Test pits would be excavated to determine the nature of the near surface materials (most likely near the toe of the landslide). Test pits will be logged, colour photographed, tested to record in situ conditions and sampled for subsequent laboratory testing. Testing will include dynamic cone penetrometer and hand penetrometer. Pits would be excavated to a depth of 3m or to machine refusal. Test pits would be backfilled with the excavation spoil on completion.
- Allowance for limited manual investigation by hand auger and DCP in specific areas of interest where access by machine is not possible.
- Detailed mapping of site features and logging of existing soil and rock exposures by a Principal Engineering Geologist.

3.3 LABORATORY TESTING

Bulk disturbed samples, thin wall samples (U63/75 tubes) and rock core would be taken for laboratory testing. A provisional laboratory test list has been included for general testing including:

- 24 Atterberg limits (soil classification).
- 24 natural moisture content (soil moisture status).
- 24 particle size distribution (soil classification).
- 12 Emerson crumb (soil erodibility – dispersion).
- 12 unconfined compressive strength – rock (rock strength).
- 3 triaxial tests on rock or clay to define (Mohr Coulomb failure parameters)
- 3 direct shear tests (peak and residual) on U63 or rock core samples of the suspected slide plane in EW claystone or clay soil layer of rock bedding plane (residual friction angle)

3.4 REPORTING

Issues addressed in the report will include the following:

- Site description including pre and post failure topography, utilizing CoN detailed survey and drone data.
- Details of fieldwork undertaken including description of surface and subsurface conditions including detailed results from surface mapping of tension cracking and ground heave.
- Presentation of factual geotechnical data including logs and laboratory testing.
- Interpretation of data and ground model.
- Soil and rock physical parameters.
- Geotechnical sections to define geotechnical site model and stability analysis geometry.
- Assessed hydrogeological conditions
- Slope stability analysis including back analysis of the May 2025 failure.

- Discussion of potential stabilisation options.
- RCA has allowed for the production of a draft report.
- A final report would be compiled following comments and consultation and provided in pdf format by email.

4 FEES AND COSTS

The scope provided is considered to represent the minimum suggested scope and may expand once an understanding of the site conditions and instability mechanism is obtained. The fee provided is for the scope described only and any additional fieldwork, laboratory testing and analysis would be additional and charged at Contract 2024/076T rates.

A breakdown of the estimated costs and fees is presented in the attached schedule at firm unit rates. We have estimated our fees and costs to be [REDACTED]

Should you wish us to undertake this work please provide a Purchase Order or Contract for the works.

Notes:

- The fee provided is for the scope described only and any additional fieldwork, laboratory testing and analysis would be additional and charged at Contract 2024/076T rates.
- A Dial Before You Dig evaluation and the cost of a locator has been costed into this proposal. We would expect details of all services present on site that might be a constraint to or at risk by the investigations would be provided.
- We understand and have assumed that existing site conditions are suitable for access by a track mounted drilling rig and excavator. Only minor allowance is made for provision of tracks and pads to drill locations.
- No allowance is made for access approvals for fieldwork. It is assumed that access will be provided.
- Works are proposed on an active landslide area. If works are shutdown or delayed due to reasons outside RCA Australia control then standby rates may need to be applied.
- No allowance has been included for professional survey of sample locations and instrumentation. It is assumed that CoN surveyor will survey test locations under RCA supervision.
- No allowance is made for traffic control and it is assumed that all test locations are accessible outside the traffic danger zone including those near Russell Road.
- We assume that CoN will provide RCA with a detailed site survey of the landslide area.
- Allowance is made for grouting the bores or portions of the bores where instrumentation is not installed.
- Allowance has been made for the disposal of excess soil generated during drilling/ test pitting works.

- No allowance is made for ongoing monitoring and analysis of groundwater and inclinometer reading. Allowance is made for setup and an initial reading. Monitoring and analysis beyond this would be at rates.
- All laboratories used for analysis are NATA accredited, unless specified otherwise, some tests may be subcontracted to other laboratories
- Our estimate does not include post-report consultations, meetings or further analysis that you may subsequently require. Post-report consultations, meetings and further analysis would be at rates.
- Where site induction is required, it would be charged at the rates shown in the attached schedule for personnel and subcontractors.
- Invoicing for this project will be progressed claimed at the end of each month for the work undertaken in that month.

We anticipate being able to commence fieldwork within one-two weeks of your instructions to proceed. Fieldwork will require about four-six weeks and laboratory testing will take about three-four weeks to complete. RCA anticipate our draft report will be available two weeks from the receipt of laboratory results.

The company is committed to Quality Assurance. The company operates under a third party certified quality assurance system complying with ISO 9001:2015 which includes our NATA registered laboratories.

Thank you again for this opportunity to work with City of Newcastle.

Yours faithfully

RCA AUSTRALIA

Dr Mark Allman
Principal Geotechnical Engineer

ATTACHMENTS

Drawing 1 – Site Topography and Test Location Plan

Drawing 2 – Mine Overlay Plan



RCA AUSTRALIA
GEO TECHNICAL • ENVIRONMENTAL

**PREFAILURE TOPOGRAPHY AND
PROPOSED BORE LOCATIONS
NEW LAMBTON LANDSLIDE
NEW LAMBTON**

CLIENT	City of Newcastle	RCA Ref	17653-101/1
DRAWN BY	JE	SCALE	1:900 (A3)
APPROVED BY	MA	DATE	25/07/2025
		DRAWING No	1
		REV	0
		OFFICE	NEWCASTLE

LEGEND

- Lot boundaries
- + Proposed bore and instrumentation locations
- Inferred surface extent of landslide ground movement based on tension crack mapping
- Approximate location of mapped rock faces
- Indicative surface slope and direction from LIDAR (red > 18°, green < 18°)
- Indicative section lines

Notes

1 Contours produced from NSW Government Spatial Services LIDAR (extracted from elevation.fstf.org.au)

0 5 10 20 30 40
metres

N



Disclaimer: 'any excerpts of original mine survey plans or record tracings and any data derived from such original mine survey plans or record tracings must not be relied upon in any way by any person, including (without limitations) for the accuracy or completeness of mine workings, and are intended for indicative purposes only. The Department of Planning and Environment is not responsible or liable to any person for any loss or liability arising out of or in connection with use of any such excerpts or derived data'

Notes:

- 1. Aerial image taken from Nearmap, 25 May 2025 (used in accordance with commercial licence)
- 2. RT682 - Borehole Seam Workings



 GEO TECHNICAL • ENVIRONMENTAL		RT682 MINE OVERLAY NEW LAMBTON LANDSLIDE	
CLIENT	City of Newcastle	RCA Ref	17653-101/1
DRAWN BY	MA	SCALE	1:2,000(A3)
APPROVED BY	MA	DATE	25/07/2025
		DRAWING No	2
		REV	0
		OFFICE	NEWCASTLE